Manual

Controlling of Shop Floor Data/Order Data

BDE-CAB 8.2

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Contents

1 Übersicht Controlling Betriebsdaten / Auftragsdaten

Purpose

Business data controlling makes it possible to evaluate data entered in the system as concerns target/actual comparisons relating to quantities, times and dates.

Implementation considerations

You use the function package if:

- You would like to evaluate the on-time delivery performance of production based on the actual order / operation deadlines
- You would like to evaluate the times and durations recorded with reference to orders
- You would like to evaluate hours accrued in an area / a department over a period of time relating to various time accounts

Integration

The business data / order data monitoring function evaluates orders or operations that were either created directly in the system or that were transferred from other systems via interfaces.

Times and quantities entered in the BDE or MDE are accessed to display order progress.

Features

- Schedule controlling
  - Tabular evaluation of delayed order and early orders
- Order statistics
  - Order statistics with evaluations of finished and active orders, articles produced with target / actual comparison of quantities, performance and times, gross / net comparison to actual performance and remaining runtime, graphic and numeric evaluations of resource performance accounts.
- Overhead cost controlling
  - Evaluation of times recorded for overhead cost orders presented in a pivot table. Individual definition of the pivot table presentation by various criteria. Graphical presentation (bar chart). Predefined detail applications for displaying times for "activities per calendar week", "cost centers per calendar week", "cost centers per activity". Graphical display of the respective results.
- Maintenance controlling

    - Evaluation of times entered for maintenance orders presented in a pivot table. Individual definition of the pivot table presentation by various criteria. Graphical presentation (bar chart). Predefined detail applications for displaying times for "articles per calendar week", "workplaces per calendar week", "articles per workplace". Graphical presentation of the respective results.
- Production controlling
    - Evaluation of times entered for production orders presented in a pivot table Individual definition of the pivot table presentation by various criteria. Graphical presentation (bar chart).
- Order profile
    - Order profile with display of times posted for operations (order duration, processing time, production and down time) incl. Gantt chart of calculated idle times.

2 Order Profile

1.1 Summary

Menu	Order Management --> Order Controlling --> Order Profile
Transaction code	orpf
Function authorization	orpf

Utilization

The order profile is indispensable for all production schedulers who want to see at a glance how orders have passed production. Due to its combination of table and Gantt chart, the production scheduler gets all required pieces of information at the push of a button.

Integration

The order profile is a report for production management

The order profile has been configured for the user to consider production orders that have already been realized. It responds to questions such as:

- Which orders have been produced (finished) within a certain period of time?
- Which output (quantities) have these orders produced?
- How much time was required?

Order controlling is made on order level and considers quantities and times with respect to the order (headers). Only finished orders are included as certain values (key figures) refer to the entire order. Consequently, the chronological selection refers to the order end.

Please note: This report is not ideal for “long-running orders” (serial orders) because of the selection option. However, the values and key figures displayed here are normally less interesting for serial orders.

Prerequisite

Only orders the order type of which is assigned to the category

- FA (production order),
- PJ (project order) or

- PM (plant maintenance order)

are displayed.

Selection Criteria

The application provides the following selection criteria:

Please take into consideration that in general - irrespective of the restrictions made using the selection criteria listed in the following – only orders that have already been started and the order type of which is assigned to the category
- FA (production order),
- PJ (project order) or
- PM (plant maintenance order)
are taken into account.

The application provides the following selection criteria:

Order

The "order" option restricts the displayed orders. Wildcards may be used.

Order type

If the order type is selected only those orders are taken into account that correspond to the selected order type. Multiple selections are possible.

Category

If the order category is selected only those orders are taken into account that correspond to the selected order type. This selection criterion refers to the category of the order type of the order. Only orders the order type of which corresponds to the specified category are displayed. Multiple selections are possible. Irrespective of this selection, only orders of the category “production order”, “project order” and “maintenance order” are taken into account.

Article

When the article is selected, all orders are determined the article number of which at the order header matches the entered value. Wildcards are taken into account.

Article designation

If the article designation is selected all orders are determined the article designation of which at the order header corresponds to the entered value. Wildcards are taken into account.

Project number

If the project number is selected all orders are determined the project number of which at the order header corresponds to the entered value. Wildcards are considered.

Sales order

If the sales order number is selected all orders are determined the sales order number of which at the order header corresponds to the entered value. Wildcards are taken into consideration.

Customer designation

When the customer designation option is selected, all orders are determined the customer designation of which at the order header matches the entered value. Wildcards are considered.

The responsibility area is not checked in this application.

"Order profile" detail application (table)

All orders matching the entered selection criteria are displayed in the “order profile” detail application.

The displayed data is described in the sections that follow (the number of columns may vary subject to the column configuration):

Status

Order status (only orders are considered that have started or that are finished)

Order

Order, article, sales order

Order header data

Target quantities

Target quantity, unit

Target quantity of the order in base quantity unit.

Actual quantities

Yield, scrap, rework, open quantity, unit

Order quantities in base quantity unit posted onto the order header. However, it is required that order backlog data that is manually created or transferred from the higher-tier system allows for the primary quantity unit to be converted into base quantity unit at operations.

Target times

Setup time

Total of setup times (setup times+additional setup time+teardown time) of all active OPs according to the order backlog. The additional setup time results dynamically from using the setup change matrix in the HYDRA shop floor scheduling module.

Processing time

Total of processing times of all active OPs according to the order backlog.

Execution time

Total of setup time and processing time according to the order backlog.

Waiting time

Total of waiting times of all active OPs according to the order backlog.

Wait time

Total of wait times of all active OPs according to the order backlog.

Transport time

Total of transport times of all active OPs according to the order backlog.

Labor utilization

Total of target labor utilization times of all active OPs. The target labor utilization time needs to be transferred via the HYD-ERP interface.

Actual times

Retention period of orders

The retention period of the order is the period between the time the order was first transferred from the PPS system ("order release" = order header creation date in HYDRA) and the time of the actual logoff (in terms of time) of the last active operation of the order.

Please note:

Whether the time of the order transfer is the time it was first transferred from the PPS system or the time it was re-transferred because it had been deleted in the meantime for technical reasons, cannot be determined or taken into consideration.

If an order is transferred more than once, and with each transfer the previous order is deleted, the creation date of the order header is the time of the last transfer.

Lead time

The order duration is the period between the time the first operation of the order is logged on and the time the last active operation is logged off

Processing time

The processing time of the order is the total of main utilization times (RPA 11) of all operations of this order.

Downtime

The downtime of the order is the total of downtimes (RPA 1 to 6, RPA 8 to 10) of all operations of the order.

Assignment time

The occupancy time is the total of setup time (RPA 7), processing time (RPA 11) and downtimes (RPA 1..6, RPA 8..10) of all operations of the order.

Wait time

The wait time results from times during which no operation was logged on. The duration is synchronized with the Gregorian calendar (shift breaks are not considered, durations posted onto RPA 12 "free break" are not considered; the posting times of the log records of the record type "U" and "E" are evaluated).

The wait time is not calculated at the time when data is requested, but by a separate process. For this purpose, however, this process needs to be integrated in the Scheduler. This procedure is described in the document entitled Activating_OrderRelatedKeyfigures.pdf.

Labor utilization

The labor utilization of the order is the total of labor utilization times of all operations of the order.

Key figures

Rate of capacity utilization

The rate of capacity utilization is the ratio of processing time (RPA 11) to occupancy/assignment time (RPA 1..11) of all operations of the order

Setup rate

The setup rate is the ratio of setup time (RPA 7) to occupancy/assignment time (RPA 1..11) of all operations of the order.

"Order profile" detail application (graphic)

The article profile shows for each operation of the order selected in the table when the individual operation was logged on.

The operation number as well as the workplace, which the operation was logged on to, are displayed on the y-axis. When it comes to split OPs, the split number is shown in addition to the operation number. Data is displayed in ascending order by the operation number.

If there are simultaneous postings for an operation (e.g. as the operation has been logged on to several machines/workplaces at the same time) data is displayed in several lines for the operation.

The postings generated in HYDRA ("ADE log") are displayed and presented as bars (Gantt chart).

The times of resource performance accounts distributed in a posting are displayed within a bar according to the corresponding colors defined for the resource performance accounts. The sequence of RPA colors (ascending by RPA number) within a posting (of a bar), however, does not describe the sequence in which RPAs are posted, but remains the same for all postings (bars).

In case the duration (the time posted onto the RPA) within the log record is less than the time between logon and logoff, the remaining time is hatched within the order profile. This can be the case if

- the OP is part of a merged operation

- the OP was posted proportionately (this might, among others, also be the case for merged OPs (MOP) logged on to the terminal)

At the end of the graphic black bars represent wait times. These times result from the time between logging the operation off and logging the next operation on (the same operation or even different operations). This means, that during that time no operation of the order was logged on.

Tooltip

When going with the mouse pointer over a bar, a window opens for the resource performance account where the mouse pointer is currently located and the following information is displayed:

- - Current operation,
- Logon/logoff time of the operation
- - Yield (in primary quantity unit) as well as quantity unit
- Duration during the logon/logoff period
- Workplace, to which the operation was logged on
- Abbreviation and designation of the resource performance account as well as the duration posted onto this resource performance account

For HYDRA-MDE machines the operation is automatically interrupted when the shift ends and automatically logged on when the shift starts. Operations that are logged on when shifts are changing get two posting records and, as a result, 2 bars are displayed in the graphic.

The sequence of RPA colors within a posting (a bar) does not represent the order in which the RPA is posted, but remains the same for all posting records (bars).

If there are simultaneous postings for an operation (e.g. as the operation has been logged on to several machines/workplaces at the same time) data is displayed in several lines for this operation.

Representation of times (italic) on the y-axis is accidental.

"Duration" detail application (graphic)

The "duration" detail application shows the below-mentioned data:

- Lead time (dark blue)
- Assignment/occupancy time (yellow)
- Processing time (light green)
- Setup time (turquoise)
- Downtime (red)
- Wait time (black)

Further information on this data can be found in the section entitled "order profile detail application (table)".

The duration is displayed to the right of each bar in the format hh:mm:ss, instead of an x-axis labeling.

3 Order-Related Statistic

Overview

	→	

Menu	Order management --> Order controlling --> Order-related statistics
Transaction code	orst
Function authorization	orst

Purpose

The order-related statistic is a detailed report about orders. You can also evaluate the quantities and times of the selected orders.

Integration

The statistic generated by the evaluation includes a list of orders and corresponding operations. The following is displayed for each operation:

- "Operations" detail application
  Tabular overview of the operations pertaining to the selected order

- “Total quantities/total times/downtimes by resource performance accounts” detail application
  Graphic overview of the quantities and times accrued on the order.

- “Performance and run time” detail application
  Tabular and graphic presentation of run times and durations based on operations

Selection criteria

The application provides the following selection criteria:

Order

You can enter the order number directly or via the search dialog.

Order type

This selection criterion refers to the order type in the order header. The application only shows the orders pertaining to the selected order type(s).

Category

This selection criterion refers to the category of the order type in the order header. The application only shows the orders pertaining to the selected category(ies).

Order status

This selection criterion refers to the order statuses in the order header. The application only shows the orders assigned to the selected order status(es).

Control

This selection criterion refers to the control indicator in the order header. The application only shows the orders assigned to the selected control indicator(s).

Article/Item

This selection criterion refers to the article in the order header. The application shows all orders that include the selected article. You can also use wild cards.

Article name

This selection criterion refers to the article name defined in the order header. The application shows all orders assigned to the selected article name. You can also use wild cards.

Sales order

This selection criterion relates to the sales order defined in the order header. The application shows all orders assigned to the selected sales order. You can also use wild cards.

Customer name

This selection criterion refers to the customer name defined in the order header. The application shows all orders including the selected customer name. You can also use wild cards.

Project number

This selection criterion refers to the project number defined in the order header. The application shows all orders of the selected project number. You can also use wild cards.

Planned order

This selection criterion refers to the planned order defined in the order header. You can also use wild cards.

Cost object

This selection criterion refers to the cost object defined in the order header. The application shows all orders of the selected cost object. You can also use wild cards.

Priority from ... to

This selection criterion refers to the priority defined in the order header. The application shows all orders assigned to the selected priority.

Basic start date from ... until

This selection criterion refers to the basic start date defined in the order header. The application only shows the orders coinciding with the selected basic start dates.

Basic end date from ... until

This selection criterion refers to the basic end date defined in the order header. The application only shows the orders coinciding with the selected basic end dates.

Order end from … until

This selection criterion refers to the actual order end defined in the order header. The application only shows orders coinciding with the selected order end dates.

Order index from ... to

This selection criterion refers to the order index defined in the order header. The application shows all orders having the selected order index.

Order group

This selection criterion refers to the order group defined in the order header. The application shows all orders that are assigned to the selected order group.

MRP controller

This selection criterion refers to the MRP controller defined in the order header. The application shows all orders of the selected MRP controller.

If you use multiple selection criteria, the order overview shows the matching results.

The responsibility area is not checked in this application.

Order overview detail application

The "order overview" detail application shows all orders you have selected in the selection panel. Every row shows one order. Numerous columns, which you may hide or show, display all available pieces of information summarized in reasonable categories.

The list shows specific order data. This includes existing order data as well as current status information on the order.

Operations detail application

The detail application *operations* shows all operations pertaining to the above selected order. If you select several orders in the order overview, this application shows all operations belonging to these selected orders. In this case, we recommend showing the "order" column.

The list shows specific operation data. This includes existing operation data as well as current status information on the operation.

Total quantities detail application

Subject to the order selected in the order overview, this detail application shows the overall collected quantities of the order. If you select several orders, the application shows the totals of the selected orders in a graphic.

Total times detail application

Subject to the order selected in the order overview, this detail application shows the overall collected times of individual operations pertaining to the order. If you select several orders, the application shows the totals of the selected orders in a graphic.

Downtimes by resource performance accounts detail application

Subject to the order selected in the order overview, this detail application shows the overall recorded downtimes of the order separated by resource performance accounts. If you select several orders, the application shows the totals of the selected orders in a graphic.

Performance and run time detail application

The “performance and run time” detail application compares the actual net and gross activities with the target activities, depending on the operations selected in the detail application “operations”.

Further information on the calculation of individual KPIs can be found here.

If you select multiple operations, the values are not added up. The detail application only shows the values of one of the selected operations.

Target/actual comparison detail application

When target/actual values are displayed and compared in a table, the target values are compared with the recorded actual values of the selected operations. The following values are displayed:

- Yield
- Scrap
- Setup time
- Cycle time
- Partitioning
- Production time per piece (time/piece)

Further information on the calculation of individual actual values can be found here.

If you select multiple operations, the values are not added up. The detail application only shows the values of one of the selected operations.

Activities detail application

The “activities” detail application compares target activities with the rendered actual activity (gross and net) of selected operations in a graphic.

Further information on the calculation of individual KPIs can be found here.

Durations detail application

The “durations” detail application compares the order duration, the production duration and downtimes of the selected operations in a graphic.

Downtimes by resource performance accounts detail application

The “downtimes by resource performance accounts” detail application compares the accrued times of the selected operations, separated onto resource performance accounts in a graphic.

Toolbar

 **Order information**

Calls up the Order information for the currently selected order.

 **Order overview**

Calls up the Order overview for the currently selected order.

 **Failure mode analysis (function authorization faep)**

Calls up the Failure Mode Analysis

 **Inspection requirement (function authorization irp)**

Calls up the application Inspection requirement

 **Inspection points (function authorization ipp)**

Calls up the application Inspection points

4 Overhead Cost Controlling

Summary

Menu	Order management → Order controlling → Overhead cost controlling
Transaction code	ohcon
Function authorization	ohcon

Usage

A portion of costs created in a company that should not be ignored are the result of so-called overhead costs. The objective of the evaluation "Overhead cost controlling" is to provide a means to make these overhead costs transparent and to identify the real "cost monsters", while in doing so finding ways to introduce countermeasures that will help lower overall costs.

Generally, it is the responsibility of the cost center managers to conduct audits and analyses of this kind and to derive the relevant measures based on the results. In production, for example, this is the responsibility of the foremen.

In addition to evaluating overhead costs relating to a specific cost center, it is also important to determine which activities incurred these costs. Overhead cost orders can be used to illustrate a breakdown of this kind.

Definition of overhead costs:

"Costs that cannot be attributed to any specific product or performance unit (cost object, cost center), such as lease or rent payments, executive salaries. [...] Overhead costs are such costs that cannot be attributed to any allocation base directly."
(Source: http://www.wirtschaftslexikon24.net/d/gemeinkosten/gemeinkosten.htm)

Integration

Database used to evaluate the order data logs .

Requirement

In order for an evaluation to be meaningful, what is required is that the employees record their overhead cost times correctly and allocate them to the cost object appropriately (correctly prepared overhead costs order).

Selection criteria

The application provides the following selection criteria:

Cost center

The postings for the record type "U"/ "E" are selected that are posted to workplaces for which the user has been authorized via the workplace's responsibility area, and which are assigned to the cost center entered.

Workplace

The postings for the record type "U"/ "E" are selected that are posted to workplace entered for which the user has been authorized via the responsibility area of the workplace.

Group

The postings for the record type "U"/ "E" are selected that are posted to workplaces for which the user has been authorized via the workplace's responsibility area, and which are assigned to the group entered.

Period

The time period entered restricts the selection by log records. Such log records are selected that have a start date within the defined period.

Responsibility area

The responsibility area that is entered restricts the requested data to the defined responsibility area for the particular machine/ workplace.

Company

By using the selection by company, only the data records for the relevant company are displayed or are included in the evaluation.

Order

The order number can be entered or selected. The order number defined here (order header number) restricts the evaluation to the selected order.

OP designation

Only the recorded data entered for an operation defined in the field OP designation with the selected designation is used for the evaluation (free text).

The responsibility area is not checked in this application.

Field descriptions

Responsibility area

The responsibility area defined at the machine for which the data was entered.

Date

The dates shown are presented broken down by day.

Workplace

By integrating the field workplace, the data can be distributed or grouped by the workplace at which the overhead costs are recorded.

Year - quarter - month - calendar week - weekday - day

This field provides the ability to group or distribute the displayed data respectively.

Group

The data shown is distributed and displayed based on the machine group that is defined at each machine/ at each workplace.

Cost center

This field allows the data to be distributed based on the cost center defined at the machine/ workplace.

Company

This field allows the data to be distributed based on the company defined at the machine/ workplace.

Order

The data shown is grouped or displayed based on the order number (order header number).

Article

Filtering by article makes it possible to filter the article number of the separate operations.

MES order number

As opposed to the order number, the MES order number is formed by combining the order number and the operation number. Therefore, the data is displayed grouped by the separate operation numbers.

OP designation

For each operation there is an operation designation that is defined in the master data at the operation.

Total duration

The sum total of all durations for the selected data records.

General detail applications

The evaluation provided in the overhead costs controlling considers the operations in the overhead costs orders category.

Target times

These relate to target times/ periods defined in the order backlog. The target times are not related proportionately to the selection period, but instead are attributed absolutely to the entire operation.

Setup time

Setup time + dismantling time + dyn. setup time

Execution times

Setup time + processing time

Actual time (definitions)

Setup time

RPA 7

Processing time

RPA 11

Execution times

Setup time + processing time

Downtime

RPA 1..6, RPA 8..10

Total duration ("Occupancy/assignment time")

Setup time + processing time + downtime time

Activities per calendar week detail applications

The data is presented in a pivot grid (top) and as a bar chart (bottom). The columns show the total duration added up per calendar week. The total durations per operation designation are presented in the lines as totals.

Cost center per calendar week detail applications

The data is presented in a pivot grid (top) and as a bar chart (bottom). The columns show the total duration added up per calendar week. The total durations per cost center are presented in the lines as totals.

Cost center per activity detail applications

The data is presented in a pivot grid (top) and as a bar chart (bottom). The columns show the total duration added up per operation designation. The total durations per cost center are presented in the lines as totals.

5 Production Controlling

Summary

Menu	Order management → Order controlling → Production controlling
Transaction code	pdcon
Function authorization	pdcon

Usage

In this evaluation, you can evaluate data from the production orders using criteria that you can define in advance.

Integration

In this evaluation, you can evaluate posting data for orders based on various criteria that can be defined in the pivot table.

Initially, the duration of the order is displayed per machine and calendar week. This display can be changed by modifying the data field combination.

Selection criteria

The application provides the following selection criteria:

Article

Restricted to articles

Article designation

Restricted to article designation

Cost center

The postings for the record type "U"/ "E" are selected that are posted to workplace for

which the user has been authorized via the workplace's responsibility area, and

which are assigned to the cost center entered.

Period

The time period entered restricts the selection by log records. The log records are selected that have a start date within the defined period.

Responsibility area

The responsibility area that is entered restricts the requested data to the defined responsibility area for the particular machine/ workplace.

Workplace

The postings for the record type "U"/ "E" are selected that are posted to the workplace entered for which the user has been authorized via the responsibility area of the workplace.

Group

The postings for the record type "U"/ "E" are selected that are posted to workplaces for which the user has been authorized via the workplace's responsibility area, and which are assigned to the group entered.

Company

By using the selection by company, only the data records for the relevant company are displayed or are included in the evaluation.

Order

The order number can be entered or selected. The order number defined here (order header number) restricts the evaluation to the selected order.

OP designation

Only the recorded data entered for an operation defined in the field OP designation with the selected designation is used for the evaluation (free text).

The responsibility area is not checked in this application.

Field descriptions

Responsibility area

The responsibility area defined at the machine for which the data was entered.

Date

The dates shown are presented broken down by day.

Workplace

By integrating the field Workplace, the data can be distributed or grouped by the workplace at which the production order costs are recorded.

Year - quarter - month - calendar week - weekday - day

This field provides the ability to group or distribute the displayed data respectively.

Group

The data shown is distributed and displayed based on the machine group that is defined at each machine/ at each workplace.

Cost center

This field allows the data to be distributed based on the cost center defined at the machine/ workplace.

Company

This field allows the data to be distributed based on the company defined at the machine/ workplace.

Order

The data shown is grouped or displayed based on the order number (order header number).

Article

Filtering by article makes it possible to filter the article number of the separate operations.

Article designation

Even if several articles have the same article number, these can be differentiated by article designation. This way the data records displayed can be grouped by article designation.

MES order number

As opposed to the order number, the MES order number is formed by combining the order number and the operation number. Therefore, the data is displayed grouped by the separate operation numbers.

OP designation

For each operation there is an operation designation that is defined in the master data at the operation.

OP

The field OP (Operation) lists the operation number only (without the order header number, e.g. 0010). This allows you to group data records which may be from different orders, however which have the same operation number.

Total duration

The sum total of all durations for the selected data records.

Tools

By running a selection by tool, only those posting records are used in the evaluation that were recorded for operations, in which the relevant tool was defined.

Pivot table detail application

You can evaluate order data based on additional criteria in the pivot view detail view.

The bar colors in the chart are set "arbitrarily" using a color chart defined internally.

6 Maintenance Controlling

Summary

Menu	Order management → Order controlling → Maintenance controlling
Transaction code	pmcon
Function authorization	pmcon

Usage

Maintenance controlling is a production management function. Maintenance especially is provided an overview of the maintenance orders that need to be processed.

The purpose of maintenance controlling is to show costs (based on activities/ times) that were incurred as a result of maintenance activities. By structuring the maintenance orders accordingly referencing the maintained equipment, the function provides the ability to identify maintenance-intensive materials.

Integration

Database used to evaluate the order data logs .

Selection criteria

The application provides the following selection criteria:

Article

Restricted to articles

Article designation

Restricted to article designation

Cost center

The postings for the record type "U"/ "E" are selected that are posted to workplaces for

which the user has been authorized via the workplace's responsibility area, and

which are assigned to the cost center entered.

Responsibility area

The responsibility area that is entered restricts the requested data to the defined responsibility area for the particular machine/ workplace.

Period

The time period entered restricts the selection by log records. Such

log records are selected that have a start date within the defined period.

Workplace

The postings for the record type "U"/ "E" are selected that are posted to workplace entered for which the user has been authorized via the responsibility area of the workplace.

Group

The postings for the record type "U"/ "E" are selected that are posted to workplaces for which the user has been authorized via the workplace's responsibility area, and which are assigned to the group entered.

Company

By running the selection by company, only the data records are displayed or used in the evaluation that were created for the machines/ workplaces at those companies that match the companies you selected.

Order

The order number can be entered or selected. The order number defined here (order header number) restricts the evaluation to the selected order.

OP designation

Only the recorded data entered for an operation defined in the field OP designation with the selected designation is used for the evaluation (free text).

The responsibility area is not checked in this application.

Field descriptions

Responsibility area

The responsibility area defined at the machine for which the data was entered.

Date

The dates shown are presented broken down by day.

Workplace

By integrating the field Workplace, the data can be distributed or grouped by the workplace at which the overhead costs are recorded.

Year - quarter - month - calendar week - weekday - day

This field provides the ability to group or distribute the displayed data respectively.

Group

The data shown is distributed and displayed based on the machine group that is defined at each machine/ at each workplace.

Cost center

This field allows the data to be distributed based on the cost center defined at the machine/ workplace.

Company

This field allows the data to be distributed based on the company defined at the machine/ workplace.

Order

The data shown is grouped or displayed based on the order number (order header number).

Article

Filtering by article makes it possible to filter the article number of the separate operations.

Article designation

Even if several articles have the same article number, these can be differentiated by article designation. This way the data records displayed can be grouped by article designation.

MES order number

As opposed to the order number, the MES order number is formed by combining the order number and the operation number. Therefore, the data is displayed grouped by the separate operation numbers.

Total duration

The sum total of all durations for the selected data records.

Tools

By running a selection by tool, only those posting records are used in the evaluation that were recorded for operations, in which the relevant tool was defined.

Article per calendar week detail applications

The data is presented in a pivot grid (top) and as a bar chart (bottom). The columns show the total duration added up per calendar week. The total durations per article number are presented in the lines as totals.

Workplace per calendar week detail applications

The data is presented in a pivot grid (top) and as a bar chart (bottom). The columns show the total duration added up per calendar week. The total durations per article number and workplace are presented in the lines as totals.

Article per work place detail applications

The data is presented in a pivot grid (top) and as a bar chart (bottom). The columns show the total duration added up per workplace. The total durations per article number are presented in the lines as totals.

7 Schedule Controlling

Overview

Menu	Order management → Order controlling → Schedule controlling
Transaction code	scec
Function authorization	scec

Available user fields		
Where?	**Object type/user field key**	**Source (type)**
Table Schedule controlling	AUNR/SYSTEM	Order (MF-D)
Table Schedule controlling	AGNR/SYSTEM	Operation (MF-D)
How to configure user fields?	Which user field types are available?	

Purpose

You use this application to monitor deviations from scheduled dates for operations. The deviations are displayed in a list. The list provides answers to the following questions, for example:

- Which operations whose planned start has been exceeded have not (yet) been started?
- Which operations are currently not being processed (status prepared or interrupted) and are expected to be delayed as a result?
- Which operations that are currently being processed (status running), are expected to be finished too late?
- Which operations were finished too late?

You can use this function to analyze lateness in the schedule, but you can also see which order start or order end is too early.

Target group for this type of information: production controllers and supervisors. Benefit of the function: The quality of the scheduled times is continuously monitored. It is a tool to ensure smooth production processes.

Integration

During the analysis, current dates are compared to planned dates. Depending on the type of planning, the function can monitor dates planned in detail, scheduled dates or basic dates. The planned dates are either the result of detailed planning performed in HYDRA shop floor scheduling or of the planning performed in the ERP system.

Requirements

To use this application properly, you must know how the planned dates are identified or set in your system. Only then you can specify a useful selection.

Selection criteria

The application provides the following selection criteria:

Check against baseline plan

The selection list shows all baseline plans that are included in the responsibility area the user is authorized for.

OPs started too early, OPs started too late, OPs finished too early, OPs finished too late

Depending on the selection options checked, the following operations are included in the selection:

- Operations started too early
- Operations finished too early
- Operations started too late
- Operations finished too late

The values that are calculated are specified by the- option selected: "Exceeding of planned dates", "Exceeding of basic dates" or "Exceeding of scheduled dates".

If none of these options is set, all operations without deviations are displayed.

Date ... to …

The system only checks the operations that have a start or end date in the specified period (depending on the selected option).

Consider long-term data

If this option is set, the system also includes operations that have been transferred from the online data to the long-term dataset.

Exceeding of planned dates

The system checks if the start and end dates specified in the detailed scheduling (e.g. HYDRA Shop Floor Scheduling) are respected.

Exceeding of basic dates

The system checks if the basic dates *Latest start* (LST) and *Latest end* (LET) transferred from the ERP system or specified in the HYDRA lead time scheduling are respected.

Note:

The ERP system must transfer the correct basic dates for the operation that result from lead time scheduling. Other option: You can also calculate the basic dates using the HYDRA lead time scheduling.

Exceeding of scheduled dates

The system checks if the calculated dates *Scheduled start time* and *Scheduled end time* transferred from the ERP system or specified in the HYDRA lead time scheduling are respected.

Planned for

Workplace or group

Workplace/group/cost center/company

Narrows down the display by workplace, group or cost center.You can also use wildcards.

Note: The selection by workplace or cost center is only useful with operations that are already planned for a specific workplace.

Order

Only operations of a specific order are selected. You can also use wildcards.

Order type

Only operations of orders of a specific order type are selected. Multiple selection is possible.

Category

Only operations of orders of a specific order type category are selected. Multiple selection is possible.

Order group

Only operations of orders of a specific order group are selected. Multiple selection is possible.

MRP controller

Only operations of orders of a specific MRP controller are selected. You can also use wildcards.

Customer name/designation

Only operations of orders for a specific customer are selected. You can also use wildcards.

Sales order

Only operations of the order that matches the specified sales order are selected. You can also use wildcards.

Project number

Only operations of the order with the specified project number are selected. You can also use wildcards.

Planned order

Only operations of the order that matches the specified planned order are selected. You can also use wildcards.

Operation status

The system only selects operations of a specific operation status. Multiple selection is possible.

Control

The system only selects operations that have a status with a specific control indicator. Multiple selection is possible.

Check responsibility area

Using this option, the user can specify if the system checks the responsibility area of the workplace or the responsibility area of the object operation/order to display data. To use this selection option, you require the function authorization chkresp.

Data collection

Depending on the option selected, the check is performed against the planned dates/the scheduled dates/the basic dates (specification as date) of the current dataset.

The values that are calculated are specified by the

- selection of the tab *Planned dates*, *Scheduled dates* or *Basic dates*.

The values *Planned start* or *Planned end* described in the tables below are used as some kind of placeholders for the different combinations (current planned dates/planned dates, current planned dates/scheduled dates, current planned dates/basic dates).

To calculate the days, the system uses the specified dates and synchronizes them with the Gregorian calendar. The displayed format is: days.hours:minutes:seconds.

Deviation from start date

Earliness

Condition 1: Control	Condition 2: Date	Calculation *)
L, U, E, A	OP actual start < planned start	ABS (OP actual start minus planned start)

Lateness

Condition 1: Control	Condition 2: Date	Calculation *)
L, U, E, A, F	OP actual start > planned start	ABS (OP actual start minus planned start)
S, V	OP planned start > today	ABS (OP planned start minus today)

On-time delivery

On-time delivery = earliness + lateness

Deviation from end date

Earliness

Condition 1: Control	Condition 2: Date	Calculation *)
E, A	OP actual end < planned end	ABS (OP actual end minus planned end)

Lateness

Condition 1: Control	Condition 2: Date	Calculation *)
S, V, L, U, F	Today > planned end	ABS (today minus planned end)
E, A	OP actual end > planned end	ABS (OP actual end minus planned end)

On-time delivery

On-time delivery = earliness + lateness

*) The time is always specified as absolute value (ABS).

Field descriptions

When the data is requested, the table shows the order/operation, OP designation and article and additionally the basic dates, the dates of detailed planning and the actual dates.

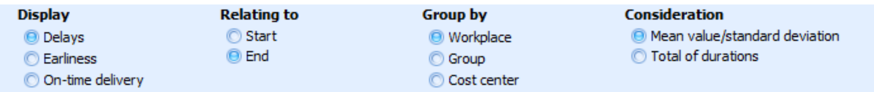
Deviations from scheduling (earliness, lateness, on-time delivery) are displayed in format days.hours:minutes:seconds.

To calculate the days, the system uses the specified dates and synchronizes them with the Gregorian calendar.

Detail application: Schedule controlling (graphic)

The detail application Schedule controlling (graphic) shows the earliness, lateness and on-time delivery for all operations selected in the table. If no operation is selected, it will be interpreted as “all operations selected”.

Display options



Display:
You can specify whether the graphic shows earliness, lateness or on-time delivery performance.

Relating to:

You can specify whether the graphic shows the deviations from the start (planned start/actual start) or from the end (planned end/actual end) of the operation.

Group by:

The selection made specifies if the graphic shows the totaled data for the workplace, group or cost center.

Consideration:

Specifies whether

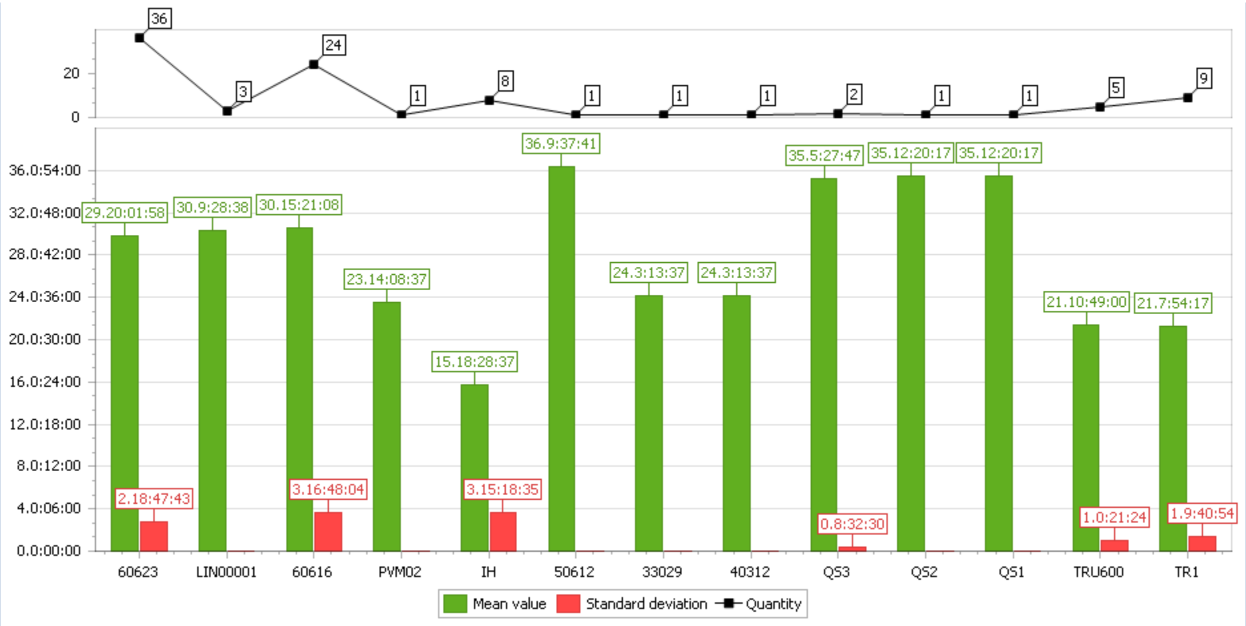
- the mean value AND the standard deviation

or

- the total of durations (lateness, …)

are displayed.

Display



Data is displayed in format Days.Hours:Minutes:Seconds.

Irrespective of the selected display options, the number of operations used for the evaluation is always displayed in form of a line.

Calculations

Mean value

The mean value is calculated as follows:

$$\mu = \frac{\sum_{x=1}^{n} Earliness_x}{n}$$

$$\mu = \frac{\sum_{x=1}^{n} Delay_x}{n}$$

$$\mu = \frac{\sum_{x=1}^{n} Earliness_x + Delay_x}{n}$$

n = Number of OPs produced too early

Standard deviation

Standard deviation is calculated as follows: $\sigma = \sqrt{Var}$

In case of lateness, the variance is calculated as follows: $Var = \frac{\sum_{x=1}^{n} (Delay_x - \mu)^2}{n}$ (analog calculation for earliness or on-time delivery).

Toolbar

The parameters to call the function or target application are generally transferred from the table. For this reason, you should always select an entry before calling an application.

 **Order information (function authorization: orin)**

Use this button to call the application Order information.

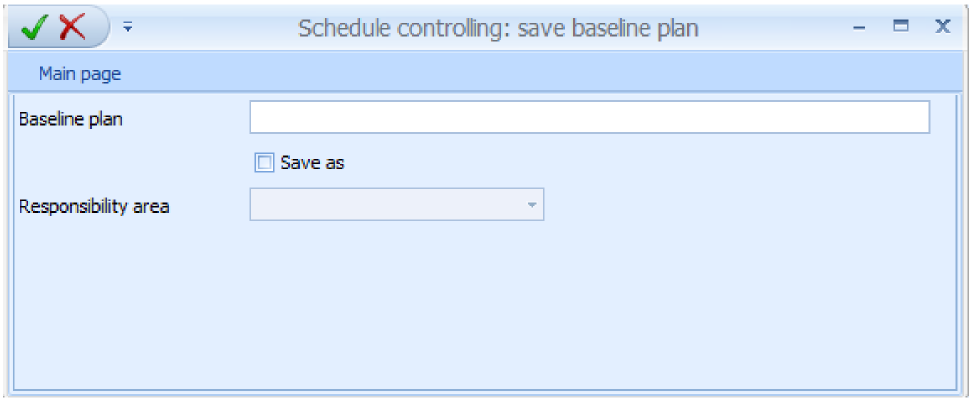
 **Order overview (function authorization: orov)**

Use this button to call the application Order overview.

Saving baseline plans

Function authorization: esvb; license: BDE-CAB

The applications *Operations* and *Pool of orders* provide this function. You can use this function to save baseline plans for selected operations that are used as selection criterion to identify deviations from planned dates.


Schedule controlling: save baseline plan
Main page
Baseline plan
Save as
Responsibility area

8 Lean Performance Analysis

Overview

Menu	Order management → Order controlling → Lean Performance Analysis
Transaction code	lpal
Function authorization	lpal

Purpose

Use the *Lean Performance Analysis* to display a value stream mapping in tabular and graphic form. The application shows orders including their operations and diverse key performance indicators (KPI).

A batch job of the HYDRA Scheduler cyclically calculates the different KPIs. The document Activating_OrderRelatedKeyfigures.pdf illustrates how you can activate this processing.

Selection criteria

If you use several selection criteria, the application *Lean Performance Analysis* shows the overlapping results.

The application provides the following selection criteria:

Order

The selection criterion *order* refers to the order number. The application shows the selected order. You can also enter wild cards.

Order type

The selection criterion *order type* refers to the order type of the order header. The application only shows the orders pertaining to the selected order type(s).

Category

The selection criterion *category* refers to the category of the order type in the order header. The application only shows the orders pertaining to the selected category(ies).

Finished article

The selection criterion *finished article* refers to the article in the order header. The application shows all orders that include the selected article. You can also use wild cards (placeholders *).

Article name

The selection criterion *article name* refers to the article name defined in the order header. The application shows all orders assigned to the selected article name. You can also use wild cards (placeholders *).

Order status

The selection criterion *order status* refers to the order statuses of the order header. The application only shows the orders assigned to the selected order status(es).

Control

The selection criterion *control* refers to the control indicator of the order header. The application only shows the orders assigned to the selected control indicator(s).

Sales order

The selection criterion *sales order* refers to the sales order defined in the order header. The application shows all orders that include this sales order. You can also use wild cards.

Customer name

The selection criterion *customer name* refers to the customer name defined in the order header. The system shows all orders including the selected customer name. You can also use wild cards (placeholders *).

Project number

This selection criterion refers to the project number defined in the order header. The application shows all orders of the selected project number. You can also use wild cards (placeholders *).

Planned order

The selection criterion *planned order* refers to the planned order defined in the order header. You can also use wild cards (placeholders *).

Cost object

The selection criterion *cost object* refers to the cost object defined in the order header. The application shows all orders of the selected cost object. You can also use wild cards (placeholders *).

Order group

- The selection criterion *order group* refers to the order group defined in the order header. The application shows all orders that are assigned to the selected order group.

Operation

The selection criterion *operation* refers to the operation number. You can also use wild cards (placeholders *).

Basic start date from ... until

The selection criterion *basic start date* refers to the basic start date defined in the order header. The application only shows the orders coinciding with the selected basic start dates.

Basic end date from ... until

The selection criterion *basic end date* refers to the basic end date defined in the order header. The application only shows the orders coinciding with the selected basic end dates.

Order end by

The selection criterion *order end* refers to the actual job end defined in the order header. The application only shows orders coinciding with the selected order end date.

Order overview detail application

Status category

Order status

The application shows the bitmap (“LED”) defined in the status configuration as the status.

By default, the color of the status LED corresponds to the color of the control LED.

Order status (text)

The status text results from the current status of the operation.

Status since

Date/time since when the order is in this status.

Order category

Shows specific data for the Orders. Relevant fields are:

Order

Number of the corresponding order

Finished article

The finished article of the complete order maintained in the order header.

Article name

Name of the finished article.

Quantities category

Target quantity (B)

Quantity specification for the order in base quantity unit.

Unit (B)

Defined unit (base quantity unit)

Yield (B)

Recorded yield in base quantity unit of the last operation that can be posted.

This is the last operation included in the order network that is neither locked (internal control flag "Y") nor deleted logically (internal control flag "D"). This operation "provides" the quantity for the entire order.

This means that the yield is 0 as long as the order has not been finished, i.e. as long as no quantity has been posted onto the order's last operation that can be posted.

If the last operation that can be posted has a quantity > 0, but the order overview does not show this quantity, check the procedure described in the document entitled Activating_OrderRelatedKeyfigures.pdf or proceed as described there.

Scrap (B)

Total of the scrap quantities entered for all operations of the order in base quantity unit. Requirement: post scrap in the base quantity unit onto the operations.

Rework (B)

Total of the rework quantities entered for all operations of the order posted in base quantity unit. Requirement: post the rework quantity in the base quantity unit onto the operations.

Open quantity (B)

Total of the open quantities entered for all operations of the order posted in the base quantity unit. Requirement: post the open quantity in the base quantity unit onto the operations.

***Target times* category**

Planned lead time

The planned lead time includes all planned execution times, like setup time, processing time, inspection time and retooling time.

Target setup time

Target setup time for the operation.

Target processing time

Target processing time for the operation.

Target execution time

Total of target setup time + target processing time

Target labor utilization

Total of the target labor utilization of all active OPs that can be posted

Target wait time

Defined target wait time of the order.

***Actual times* category**

Retention period of order

The retention period of the order results from the period of time between:

- when the order was first transferred from the ERP system ("order release" = creation date of the order header in HYDRA) and
- when the last (chronological) active operation of the order is actually logged off.

Note: We cannot identify or take into account if

- the ERP system transfers the order for the first time or
- the order has meanwhile been deleted and resent due to technical problems.

Lead time

The order duration results from the period of time between:

- the first logon of an operation of the order and
- the logoff of the last active operation (from a chronological view).

Setup time

The setup time of the order is the total of setup times (RPA 7) of all active operations that can be posted.

Processing time

The processing time of the order is the total of main production times (RPA 11) of all active operations that can be posted.

Downtime

The downtime of the order is the total of downtimes (RPA 1 to 6, RPA 8 to 10) of all active operations that can be posted.

Occupancy time

The assignment/occupancy time is the total of the setup times (RPA 7), processing times (RPA 11) and downtimes (RPA 1...6, RPA 8...10) of all active operations that can be posted.

Personnel deployment/labor utilization

The labor utilization of the order is the total of personnel deployment times of all active operations that can be posted.

***Key performance indicators* category**

Utilization efficiency (rate of capacity utilization)

The capacity utilization rate/utilization efficiency is the ratio of the processing time (RPA 11) to the assignment/occupancy time (RPA 1 ... 11) in percent.

$$Utilization\ efficiency = \frac{\mathrm{RPA11}}{\sum_{n=1}^{11} RPAn} * 100$$

Setup ratio

The setup ratio is the ratio of the setup time/costs (RPA 7) to the assignment/occupancy time (RPA 1...11) in percent.

***RPA* category**

RPA 1-12

Actual times posted onto the resource performance accounts (RPA).

***Deviation* category**

Yield/target quantity [%]

Comparison of the produced yield (B) and the planned target quantity (B) in %.

$$\frac{\text{Yield (B)}}{T\arg et\ quantity(B)}$$

Processing [%]

Comparison of the target processing time and the processing time actually posted onto RPA 11 in %.

$$\frac{\text{Processing time}}{T\arg et\ proces\sin g\ time}$$

Setup [%]

Total (100 / target setup time * setup time) of all finished operations that can be posted.

Labor utilization [%]

Total (100 / target labor utilization * labor utilization) of all finished operations that can be posted. Only if the labor utilization > 0. You must specify the target labor utilization/personnel deployment if you want to calculate this value.

Operations detail application

The detail application *operations* shows all operations pertaining to the above selected order.

If you select several orders in the order overview, this application shows all operations belonging to these selected orders. In this case, we recommend showing the "order" column.

***Operation* category**

Shows specific data for the operations.

***Status* category**

Status

This category shows the bitmap (“LED”) defined in the status configuration as the operation status.

By default, the color of the status LED corresponds to the color of the control LED.

Status text

The status text results from the current status of the operation.

Status since

Point in time since the status is available.

The field is empty for prepared operations.

Predecessor status

Status of the predecessor operation. This status indicates whether the predecessor operation has already been started and thus material, which will be further processed in the current operation, has already been processed or produced.

***Primary quantity/secondary quantity/tertiary quantity* category**

Target quantity

Quantity specifications for the Operation.

Yield

The *yield* column shows the recorded yield quantity.

Scrap

The *scrap* column shows the recorded scrap quantity.

Rework

Quantity that has to be reworked.

Open quantity

The *open quantity* is another quantity account.

Unit

Quantity unit of the displayed values.

The quantities listed here are displayed as base, primary, secondary and tertiary quantity. In the majority of cases, we recommend displaying only one of these quantity types. The terminal collects quantities in the *primary quantity.*

***Key performance indicators* category**

Actual cycle

Use the following formula to calculate this actual cycle:

$$\frac{\text{RPA 11}}{(Yield_{\text{Primary quantity}} / Partitioning)}$$

Setup time

Total of the operation times posted onto the resource performance accounts 7.

Processing time

Total of the operation times posted onto the resource performance accounts 11.

Utilization efficiency (rate of capacity utilization)

The capacity utilization rate/utilization efficiency is the ratio of the processing time (RPA 11) to the assignment/occupancy time (RPA 1 ... 11) in percent.

$$Utilization\,efficiency = \frac{\mathrm{RPA11}}{\sum_{n=1}^{11} RPAn} * 100$$

Setup ratio [%]

The setup ratio is the ratio of the setup time/costs (RPA 7) to the assignment/occupancy time (RPA 1...11) in percent.

$$Setup\ ratio = \frac{\mathrm{RPA7}}{\sum_{n=1}^{11} RPAn} * 100$$

Total gross duration

Total of the operation times posted onto the resource performance accounts 1-12.

Yield, scrap

Produced yield and/or scrap quantity in the respective primary, secondary, tertiary or base quantity unit.

Lead time

The lead time results from the period of time between:

- the first logon of the operation and
- the last logoff of the operation (if the operation is finished) or
- the last interruption (if the operation is not finished). Lead times are synchronized based on the Gregorian calendar (the shift calendar is not taken into account). The lead time is 0 if no difference can be calculated (between logging on and off).

Wait time

The wait time includes those times of the lead time during which the operation was not logged on (lead time minus RPA 1-12). Times are calculated based on the BDE log records. The wait times are synchronized with the Gregorian calendar.

Execution time

Total of the times posted onto resource performance account 7 and 11.

Downtime

Total of the times posted onto the resource performance accounts 1 - 6 and 8-10.

Lean Performance Analysis detail application (graphic)

The graphic shows the key performance indicators for an order and its operations. The graphic aligns the operations according to their order/operation number. Use the dialog "settings" to specify the KPIs you want to display.

The following information is shown for operations:

- MES order number
- OP name
- Article/Item
- Article name
- The selected KPIs (see above for the calculation *"category: key performance indicators")*

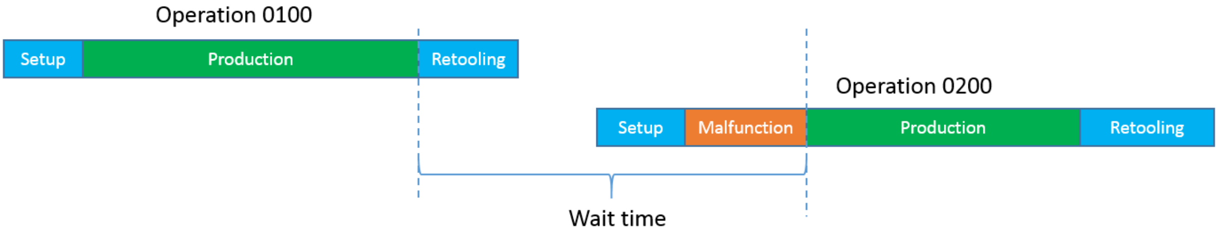
You can also view the wait time and/or the transition time between operations:

Transition time (between OPs)

The transition time of an operation refers to the time period between
- the (last) actual logoff of the current ("preceding") operation and
- the (first) actual logon of the next ("subsequent") operation. The transition time is 0 if two OPs coincide. Times are synchronized with the Gregorian calendar.

Wait time (between OPs)

Period of time starting:
- when the preceding operation is no longer in the status "production" and
- the subsequent operation starts producing (status "production")..



Times are synchronized with the Gregorian calendar.

The system can only calculate the KPI, once:
- the preceding operation has been logged off and
- the next operation has been logged on.

The system does not recalculate the KPI if you edit the MDE postings subsequently.

If, contrary to expectations, the KPI is 0, check the procedure described in the document entitled [Activating_OrderRelatedKeyfigures.pdf] or proceed as described there.

The application shows the KPIs for each operation and order:

Toolbar

 **Order information (function authorization: orin)**

Calls up the Order information

 **Order overview (function authorization: orov)**

Calls up the Order overview

 **Settings**

Here you can choose from the KPIs you want to display for the orders and operations.